

A Comprehensive Simulation of the UEFA Champions League

Allen Roman

Department of Computer Science
Kennesaw State University
Kennesaw, GA, USA
aroman14@students.kennesaw.edu

Abstract—This paper presents a comprehensive simulation of the UEFA Champions League (UCL) tournament using discrete-event simulation and machine learning to model player and team behavior. The simulation incorporates detailed player statistics, team formations, and tactical styles to create a realistic representation of football matches. By simulating matches at a second-by-second resolution, the model captures key events such as passes, shots, tackles, and goals while accounting for dynamic factors like player fatigue, team tactics, and match context. At the core of the simulation are two complementary machine learning components: the Large Events Model (LEM) for predicting on-ball events and a transformer-based Off-Ball Movement Predictor for modeling player positioning. Our validation process demonstrates that the simulation produces statistically plausible results comparable to real UCL matches, successfully replicating emergent patterns in team performance and providing insights into the factors that influence match outcomes. This simulation framework offers a valuable tool for analyzing tactical approaches, evaluating player performance under various conditions, and exploring hypothetical scenarios in professional football.

Index Terms—soccer simulation, discrete-event simulation, machine learning, transformer models, sports analytics, football, tactical analysis

I. INTRODUCTION

The UEFA Champions League is the premier club football competition in Europe, featuring elite teams competing in a tournament format that includes both group stages and knockout rounds. This project aims to create a realistic simulation of the entire tournament, modeling both individual matches and the overall competition structure.

The simulation combines discrete-event simulation techniques with machine learning models to create a detailed representation of soccer matches. It accounts for real-world details including player statistics, team formations, and playing styles while modeling the match dynamics on a second-by-second basis. This granular approach allows for capturing major events like passes, tackles, shots, and substitutions with high fidelity.

A. Objectives and Motivation

The primary objectives of this simulation project are:

- To create a realistic model of UCL matches that accurately reflects player and team behavior
- To analyze how different factors such as team strategies, player attributes, and match conditions affect outcomes

- To provide a framework for testing tactical hypotheses and strategic scenarios
- To simulate complete UCL tournaments, including group stages, knockout rounds, and final matches

The motivation for this work stems from the growing importance of data-driven analysis in professional football. As teams increasingly rely on advanced analytics to gain competitive advantages, there is a need for sophisticated simulation tools that can model the complex dynamics of football matches. This project contributes to the field of sports analytics by demonstrating how computational methods can be applied to understand and predict match outcomes.

B. Research Questions

The simulation aims to answer several key questions about football matches:

- How do team strategies (e.g., possession-based play vs. counter-attacking) influence match outcomes?
- What impact do player attributes have on team performance and match dynamics?
- How do dynamic factors such as fatigue, injuries, and tactical adjustments affect matches?
- What patterns emerge from simulating complete UCL tournaments, and how do they compare to real-world results?

By addressing these questions, the simulation provides insights that could be valuable for coaches, analysts, and football enthusiasts alike.

II. MACHINE LEARNING MODELS

A. Large Events Model (LEM)

The Large Events Model (LEM) is one of the core components of our simulation, responsible for predicting on-ball events that drive the match narrative. The LEM is a deep learning model that predicts events such as passes, shots, tackles, and goals based on the current match state. It takes into account:

- Current match context (time, score, player positions)
- Recent match history (previous 5 events)
- Player attributes (passing accuracy, shooting power, etc.)
- Team tactical setups

The LEM implemented in our system is a modified version of the approach in the `nvsclub/LargeEventsModel` repository. For each time step, the LEM predicts:

- Event type (e.g., PASS, SHOT, TACKLE, DRIBBLE)
- Spatial coordinates for the event (start and end positions)
- Probability of success for the action

Mathematically, the LEM can be represented as a function:

$$E_{t+1}, P_{success} = f_{LEM}(S_t, H_t, A_{player}, T_{tactics}) \quad (1)$$

Where E_{t+1} is the next event, $P_{success}$ is the probability of success, S_t is the current state, H_t is the recent event history, A_{player} is the player attributes, and $T_{tactics}$ represents team tactical parameters.

The LEM uses a multi-head architecture with specialized components for:

- Event type prediction (classification task)
- Spatial coordinates prediction (regression task)
- Success probability estimation

This design allows the model to capture the complex relationships between these different aspects of on-ball events.

B. Off-Ball Movement Predictor

The second key machine learning component is the Off-Ball Movement Predictor, which models how players without the ball position themselves. This is crucial for creating realistic match dynamics, as most players at any given moment are not directly involved with the ball.

Our implementation uses a transformer-based architecture that predicts player movements based on:

- Current player positions relative to the ball
- Team tactical styles
- Player attributes
- Ball possession state

The model architecture includes:

- Input embedding layers for player features, team styles, and positions
- Multi-head attention mechanisms to capture player-to-player interactions
- Position-specific movement patterns

The forward pass of the model can be represented as:

$$P_{t+1} = f_{movement}(P_t, B_t, A_{player}, T_{style}, M_{possession}) \quad (2)$$

Where P_{t+1} are the predicted player positions, P_t are current positions, B_t is the ball position, A_{player} are player attributes, T_{style} represents team tactical styles, and $M_{possession}$ is a mask indicating which team has possession.

The Off-Ball Movement Predictor is implemented as a PyTorch module with a transformer encoder that processes embeddings of player attributes, relative positions, and team tactical features:

$$\begin{aligned} E_{player} &= \text{PlayerEmbedding}(A_{player}) \\ E_{position} &= \text{PositionEmbedding}(P_t - B_t) \\ E_{team} &= \text{TeamEmbedding}(T_{style}) \\ E_{combined} &= E_{player} + E_{position} + E_{team} \\ H &= \text{TransformerEncoder}(E_{combined}) \\ \Delta P &= \text{OutputLayer}(H) \\ P_{t+1} &= P_t + \Delta P \end{aligned} \quad (3)$$

This architecture enables the model to capture the complex interdependencies between players' positions and movements, creating emergent team behaviors such as defensive compactness, attacking width, and tactical formations.

III. SYSTEM ARCHITECTURE

A. State Representation Module

The state representation module contains classes that define and maintain the current state of all entities in the simulation:

1) *Ball Class*: The Ball class encapsulates all aspects of the match ball:

- Physical State: Position coordinates (x,y), height above ground, velocity vector, and speed
- Status Attributes: Current physical state (on ground, in air, stationary), action being performed (passed, shot, loose), and field zone location
- Possession Information: References to the team and player currently in possession

2) *Player Class*: The Player class represents individual football players, the primary actors in the simulation:

- Identity: Player ID, name, assigned position, and team affiliation
- Attributes: Performance capabilities (passing, shooting, defending, stamina, etc.) that affect action success rates
- Physical State: Position, orientation, velocity, speed, and acceleration on the pitch
- Action State: Current action being performed, action phase (starting, executing, finishing), action target, and availability for new actions
- Physiological State: Current stamina, fatigue level, injury status, and sprint availability

3) *Team Class*: The Team class manages groups of players and team-level information:

- Identity: Team ID and name
- Match State: Possession status, goals scored, and goals conceded
- Player Management: Collections of all players, active lineup, and substitutes
- Tactical Setup: Formation, phase of play, and tactical parameters (pressing intensity, defensive line height, width, tempo, etc.)

4) *Match Class*: The Match class serves as the container for the entire simulation:

- Core Components: References to the competing teams and the ball
- Match State: Current clock time, match period, game phase, and set piece status
- Possession Tracking: Reference to the team currently in possession
- Match History: Chronological record of all significant events

B. Simulation Engine Module

The IntegratedMatchSimulator class drives the entire simulation using SimPy's discrete-event framework:

- Initialization: Sets up the match with teams, players, and initial conditions
- Time Management: Advances time in one-second increments
- Process Coordination: Manages the main match process and coordinates player processes
- State Updates: Updates all player positions, actions, and stamina levels each second
- Event Processing: Detects and handles events like goals, fouls, and out-of-bounds

C. Enhanced Movement System

While the Off-Ball Movement Predictor provides a machine learning approach to player movement, we also implemented an alternative rule-based system called EnhancedMovement. This system considers:

- Formation Templates: Normalized positions for common formations (4-4-2, 4-3-3, etc.) in both attacking and defending phases
- Team Tactical Styles: Consideration of possession style, defensive approach, width, and attacking mindset
- Player Attributes: Individual characteristics affecting movement patterns and decision-making
- Territorial Advantage: Quality differences between teams affecting field position

The system calculates appropriate movements for different player roles:

- Defenders track attackers near the goal in defensive scenarios
- Midfielders provide passing options and create space
- Attackers make intelligent runs into the box when the team is attacking in the final third

D. Tournament Simulation

The ChampionsLeagueTournament class simulates a complete UEFA Champions League tournament:

- Group Stage: Creates 8 groups of 4 teams based on seeding, with teams drawn from different pots
- Round-Robin Matches: Simulates all group stage matches and calculates standings
- Knockout Stage: Models the Round of 16, Quarter-finals, Semi-finals, and Final

- Match Resolution: Handles matches including extra time and penalty shootouts when needed

IV. IMPLEMENTATION

A. Data Collection and Processing

The simulation relies on comprehensive data about teams and players:

- Player Attributes: Technical skills (passing, shooting), physical characteristics (pace, stamina), and positional tendencies
- Team Tactics: Formations, playing styles, and tactical parameters
- Historical Event Data: For training the LEM and validating outputs

Data was collected using web scraping techniques from sources like FBref and WhoScored, with appropriate cleaning and normalization processes applied.

B. Discrete-Event Simulation Implementation

The simulation engine was implemented using Python and the SimPy library. The engine advances time in one-second increments, with each step involving:

- Predicting the next event using the LEM or a probabilistic fallback
- Updating player positions using the Off-Ball Movement Predictor or EnhancedMovement system
- Processing event outcomes and updating match state
- Recording events and position history for later analysis

Algorithm 1 Match Simulation Algorithm

```

RunMatch(homeTeam, awayTeam, duration)
Initialize player positions based on formations
Set initial ball position and possession time ← 0
while time < duration
do
    3: event ← PredictNextEvent()
    6: HandleEvent(event)
    7: UpdatePlayerPositions()
    8: RecordHistory()
    9: time ← time + interval
10: end while
11: return match results

```

C. Machine Learning Implementation

1) *LEM Implementation*: The LEM was implemented using PyTorch, with a multi-head architecture:

- Shared encoder layers to process match state
- Specialized heads for event type, spatial coordinates, and outcome prediction
- Training on historical event data with data augmentation techniques

The event prediction process follows these steps:

Algorithm 2 LEM Event Prediction

```

PredictNextEventmatchState, history, playerAttributes
encodedState  $\leftarrow$  EncodeMatchState(matchState, history)
eventTypeProbs  $\leftarrow$  EventTypeHead(encodedState)
eventType  $\leftarrow$  SampleFromDistribution(eventTypeProbs)
spatialParams  $\leftarrow$  SpatialHead(encodedState, eventType)
successProb  $\leftarrow$  OutcomeHead(encodedState, eventType,
spatialParams) return event, successProb

```

2) *Off-Ball Movement Implementation*: The Off-Ball Movement Predictor was implemented as a transformer-based neural network in PyTorch:

- Input embedding layers for player features, team styles, and positions
- Transformer encoder with multi-head attention for modeling player interdependencies
- Output layer for predicting position adjustments

Algorithm 3 Off-Ball Movement Prediction

```

PredictMovementsplayers, positions, ballPosition, pos-
session playerFeatures  $\leftarrow$  ExtractPlayerFeatures(players)
teamFeatures  $\leftarrow$  ExtractTeamFeatures(teams) relPositions
 $\leftarrow$  positions - ballPosition playerEmbeddings  $\leftarrow$ 
PlayerEmbedding(playerFeatures) positionEmbeddings
 $\leftarrow$  PositionEmbedding(relPositions) teamEmbeddings
 $\leftarrow$  TeamEmbedding(teamFeatures, possession)
combinedEmbeddings  $\leftarrow$  playerEmbeddings + position-
Embeddings + teamEmbeddings transformerOutput
 $\leftarrow$  TransformerEncoder(combinedEmbeddings)
positionAdjustments  $\leftarrow$  OutputLayer(transformerOutput)
newPositions  $\leftarrow$  positions + positionAdjustments return
newPositions

```

V. RESULTS AND ANALYSIS

A. Match Simulation Outcomes

Our simulation generated results that align with real-world patterns observed in UEFA Champions League matches. Key findings include:

1) *Match Statistics*: Statistical validation revealed that our simulation produces match statistics within acceptable margins of error compared to historical UCL data:

TABLE I
COMPARISON OF SIMULATED VS. HISTORICAL MATCH STATISTICS

Metric	Simulation	Historical	Difference
Goals per match	2.5	2.8	-10.7%
Home team wins	44%	47%	-6.4%
Away team wins	32%	28%	+14.3%
Draws	24%	25%	-4.0%

These differences fall within our predefined acceptable margins, indicating that the simulation captures the fundamental dynamics of UCL matches reasonably well. The slightly higher rate of away team success in our simulation may be due to our simplified home advantage implementation.

2) *Player Movement Analysis*: Cross-model validation comparing our enhanced movement model with a neural network approach showed:

TABLE II
COMPARISON OF PLAYER MOVEMENT MODELS

Metric	Enhanced	Neural	Difference
Total player movement	1250m	1320m	+5.6%
Team positioning accuracy	82%	78%	-4.9%
Tactical formation maintenance	85%	75%	-11.8%
Response to ball position	90%	85%	-5.6%

The enhanced movement model demonstrated better tactical positioning and formation maintenance, providing more consistent and realistic player movements during matches.

3) *Team Tactical Analysis*: When testing different tactical setups, we observed significant variations in performance metrics:

TABLE III
TEAM PERFORMANCE ACROSS DIFFERENT TACTICAL APPROACHES

Metric	Baseline	High-Pressing	Counter-Attack	Uneven
Goals Scored	1.48	1.92	1.12	2.38
Goals Conceded	1.48	1.73	0.89	0.76
Possession %	50.2%	58.3%	41.7%	64.3%
Win %	33.0%	41.0%	37.0%	72.0%

These results demonstrate that the simulation effectively captures the trade-offs inherent in different tactical approaches. The high-pressing approach generates more offensive opportunities but leaves teams more vulnerable defensively, while the counter-attacking style sacrifices possession for defensive solidity.

B. Tournament Simulation Analysis

Our tournament simulations successfully modeled the complete UCL format, including:

- Group stage with 8 groups of 4 teams
- Round of 16 with group winners facing runners-up
- Quarter-finals, Semi-finals, and Final matches
- Handling of tiebreakers, extra time, and penalty shootouts

When team quality was varied, significant differences in tournament outcomes emerged:

- Teams with higher quality ratings (90) dominated matches against lower-rated teams (70), achieving 72% win rates
- Quality differentials outweighed tactical advantages in most matchups
- Home advantage provided a modest benefit but was secondary to team quality

C. Sensitivity Analysis

Our sensitivity analysis revealed which factors most significantly influence match outcomes:

- 1) Tactical Parameters:

- Attacking Mindset showed the highest impact (elasticity of 1.2)
- Defensive Style significantly affected goal-conceding rates
- Formations influenced both offensive and defensive performance

2) Player Attributes:

- Finishing ability demonstrated the highest impact on goal-scoring
- Pace showed moderate influence on overall performance
- Tackling significantly affected duels won but had limited impact on goal differential

3) Model Parameters:

- Shot success probability had the most direct impact on match outcomes
- Pass success probability influenced possession metrics but had less direct impact on scoring
- Duel outcome weights showed minimal effect on overall results

These findings highlight that tactical decisions and key player attributes (particularly finishing) have the greatest influence on match outcomes, while many model parameters have comparatively minor effects.

VI. DISCUSSION

A. Key Findings and Implications

Our comprehensive simulation of the UEFA Champions League yielded several significant findings:

1) *Tactical Parameters Dominate:* Team tactical setups, particularly attacking mindset, defensive style, and formation, have the most substantial impact on match outcomes. This suggests that coaches should prioritize tactical coherence and appropriate strategic approaches for different opponents.

2) *Finishing Quality Matters Most:* Among player attributes, finishing ability demonstrates the highest influence on goal-scoring outcomes. This aligns with football wisdom that clinical finishing is often the difference between winning and losing close matches.

3) *Possession Not Determinative:* High possession does not guarantee success; defensive counter-attacking approaches can achieve comparable results with less possession. This reflects real-world examples of successful counter-attacking teams in the Champions League.

4) *Team Quality Gap Is Decisive:* When team quality differs significantly, tactical choices become secondary factors in determining match outcomes. This explains why underdogs in the Champions League often need to employ specific tactical approaches to overcome quality disadvantages.

B. Strengths and Limitations

1) *Strengths:* The simulation demonstrates several key strengths:

- Realistic tournament structure and progression
- Statistically reasonable match outcomes

- Robust handling of edge cases
- Efficient integration of player and team data

The simulation's discrete-event approach allows for detailed tracking of match dynamics with second-by-second precision. The integration of machine learning models enables more sophisticated decision-making than traditional statistical methods.

2) *Limitations:* Our model also has several limitations:

- Simplified Team Tactics: Basic tactical representations don't capture the full complexity of real soccer team strategies
- Limited Player Individuality: While player attributes affect movement and performance, the model doesn't capture unique player styles or specialized roles
- Basic Physics: Simplified physics doesn't account for factors like weather, fatigue, or pitch conditions
- Randomness vs. Skill: The balance between randomness and skill-based outcomes could be further refined
- Match Context: The model doesn't account for tournament context effects like risk-taking in must-win matches

C. Future Research Directions

Based on our analysis, several promising directions for future research emerge:

1) Advanced Modeling Approaches:

- Development of more sophisticated multi-agent reinforcement learning models
- Integration of psychological and tactical factors into simulation models
- Improved methods for real-time analysis and prediction
- Enhanced frameworks for modeling player interactions and team dynamics

2) Data Integration and Analysis:

- Better methods for combining different types of match data
- Improved approaches for handling missing or noisy data
- Development of standardized data formats and analysis protocols
- Enhanced techniques for real-time data processing

3) Practical Applications:

- Development of more user-friendly analysis tools for coaches and analysts
- Creation of automated tactical recommendation systems
- Improved injury prediction and prevention systems
- Enhanced methods for player development and scouting

Bridging the gap between real-world and virtual domains represents a particularly promising direction for future research. This includes developing better simulation environments and improving the transfer of learned strategies from virtual to real-world scenarios.

VII. CONCLUSION

This project has successfully developed a comprehensive simulation of the UEFA Champions League using discrete-event simulation and machine learning techniques. By model-

ing matches at a second-by-second resolution and incorporating detailed player and team attributes, the simulation creates a realistic representation of football dynamics that aligns with statistical patterns observed in real UCL matches.

The integration of the Large Events Model (LEM) for predicting on-ball actions and the transformer-based Off-Ball Movement Predictor represents a significant advancement over traditional statistical approaches. These machine learning components enable more dynamic and context-aware simulation of football matches, with the LEM predicting event types, spatial coordinates, and success probabilities while the movement model captures the complex interdependencies between player positions.

All critical issues identified during our validation process have been addressed, and the model now produces statistically plausible match results within acceptable margins of error compared to real-world data. The simulation successfully replicates key aspects of football matches, including the trade-offs between different tactical approaches and the impact of player attributes on performance.

As sports analytics continues to evolve, this type of simulation framework will become increasingly valuable for teams, analysts, and researchers seeking to understand the complex factors that influence football performance and tournament outcomes. Future work can build upon this foundation to create even more sophisticated models that capture additional aspects of football dynamics and provide deeper insights into the beautiful game.

REFERENCES

- [1] F. Dambroz, F. M. Clemente, and I. Teoldo, "The effect of physical fatigue on the performance of soccer players: A systematic review," *PLOS ONE*, vol. 17, no. 7, pp. e0270099, 2022.
- [2] M. Konefał et al., "The seven phases of match status differentiate the running performance of soccer players in UEFA Champions League," *Scientific Reports*, vol. 13, pp. 6675, 2023.
- [3] G. P. Nassis, E. Verhagen, J. Brito, P. Figueiredo, and P. Krstrup, "A review of machine learning applications in soccer with an emphasis on injury risk," *Biology of Sport*, vol. 40, no. 1, pp. 233–239, 2023.
- [4] Z. Pu et al., "Orientation and decision-making for soccer based on sports analytics and AI: A systematic review," *IEEE/CAA Journal of Automatica Sinica*, vol. 11, no. 1, pp. 37–57, 2024.
- [5] M. Rico-González, J. Pino-Ortega, A. Méndez, F. M. Clemente, and A. Baca, "Machine learning application in soccer: a systematic review," *Biology of Sport*, vol. 40, no. 1, pp. 249–263, 2023.
- [6] H. Rue and Ø. Salvesen, "Prediction and retrospective analysis of soccer matches in a league," *Journal of the Royal Statistical Society: Series D (The Statistician)*, vol. 49, no. 3, pp. 399–418, 2000.
- [7] A. Wong, E. Li, H. Le, G. Bhangu, and S. Bhatia, "A predictive analytics framework for forecasting soccer match outcomes using machine learning models," *Decision Analytics Journal*, vol. 14, pp. 100537, 2025.